MATH 411: Honors Complex Variables

Midterm 2 Review Problems

Fall 2026 | Midterm 2: Friday, November 6, 2026, 3:00–3:50 PM, CAB 369

These problems are for practice. They cover Lectures 13–23 (Lang §§3.3–3.7, 4.1), as described on the Midterm 2 review sheet. The exam is closed book, so practise writing complete solutions without notes, stating each result you use and checking its hypotheses.

1. (Cauchy estimates and growth.)

(a) State the Cauchy estimates for $f^{(n)}(z_0)$, including the hypotheses.

(b) Let f be entire with $|f(z)| \le 2 + 3|z|^{5/2}$ for all $z \in \mathbb{C}$. Prove that f is a polynomial of degree at most 2.

(c) Is there an entire function with $|f(z)| \le 1 + |z|^{1/2}$ for all z and $f(i) \neq f(0)$?

2. (Liouville's theorem.)

(a) Let f be entire with $\operatorname{Re} f(z) \le 4$ for all z. Prove that f is constant.

(b) Let f be entire with $f(z+1) = f(z)$ and $f(z+i) = f(z)$ for all z. Prove that f is constant.

(c) Use Liouville's theorem to prove that every nonconstant polynomial has a root in $\mathbb{C}$.

3. (Laurent series on different annuli.) Let $f(z) = \dfrac{1}{z(z-2)}$.

(a) Find the Laurent expansion of f on $0 < |z| < 2$.

(b) Find the Laurent expansion of f on $|z| > 2$.

(c) Compute $\int_{|z|=1} f(z)\,dz$ and $\int_{|z|=3} f(z)\,dz$ (counterclockwise) using (a) and (b).

4. (More Laurent series.) Compute the indicated Laurent expansion and use it to evaluate the integral (all circles counterclockwise).

(a) $z^2 e^{1/z}$ on $0 < |z| < \infty$; $\displaystyle\int_{|z|=1} z^2 e^{1/z}\,dz$.

(b) $\dfrac{1}{z^2(1-z)}$ on $0 < |z| < 1$; $\displaystyle\int_{|z|=1/2} \frac{dz}{z^2(1-z)}$.

(c) $\dfrac{1}{z^2(1-z)}$ on $|z| > 1$; $\displaystyle\int_{|z|=2} \frac{dz}{z^2(1-z)}$.

5. (Logarithms and roots.)

(a) Let U be simply connected and f holomorphic and nowhere zero on U. Prove that there is a holomorphic g on U with $g^3 = f$.

(b) Prove that there is no continuous $g : \mathbb{C} \setminus \{0\} \to \mathbb{C}$ with $g(z)^3 = z$.

(c) With Log the principal branch, compute $\operatorname{Log}((-1+i)^2)$ and $2\operatorname{Log}(-1+i)$. Explain the discrepancy.

6. (Shifting a rectangular contour.) Let $a > 0$ and $b \in \mathbb{R}$. Using $\int_{-\infty}^{\infty} e^{-ax^2}\,dx = \sqrt{\pi/a}$, prove that

$$\int_{-\infty}^{\infty} e^{-ax^2}\cos(bx)\,dx = \sqrt{\frac{\pi}{a}}\,e^{-b^2/(4a)}.$$

Hint: integrate e^{-az^2} around the rectangle with vertices $\pm R$, $\pm R + ic$ for a suitable c.

7. (A sector contour: Fresnel integrals.) Integrate e^{iz^2} over the boundary of the sector $\{re^{i\theta} : 0 \le r \le R,\ 0 \le \theta \le \pi/4\}$ to prove that

$$\int_0^{\infty} \cos(t^2)\,dt = \int_0^{\infty} \sin(t^2)\,dt = \frac{\sqrt{2\pi}}{4}.$$

You may use $\int_0^{\infty} e^{-r^2}dr = \frac{\sqrt{\pi}}{2}$ and the inequality $\sin 2\theta \ge \frac{4\theta}{\pi}$ for $0 \le \theta \le \frac{\pi}{4}$.

8. (An indented semicircle.) Prove that $\lim_{R\to\infty} \int_0^R \frac{\sin x}{x}\,dx = \frac{\pi}{2}$ by integrating e^{iz}/z over the closed contour made of $[-R, -\varepsilon]$, the upper semicircle $|z| = \varepsilon$ (clockwise), $[\varepsilon, R]$, and the upper semicircle C_R of radius R (counterclockwise). *Hints:* write $\frac{e^{iz}}{z} = \frac{1}{z} + E(z)$ with E entire, and use $\sin\theta \ge \frac{2\theta}{\pi}$ on $[0, \frac{\pi}{2}]$.

9. (Cauchy's integral formula.) Compute, with all circles counterclockwise:

(a) $\displaystyle\int_{|z|=2} \frac{e^z}{(z-1)^2}\,dz$

(b) $\displaystyle\int_{|z|=2} \frac{\sin z}{z^4}\,dz$

(c) $\displaystyle\int_{|z|=1} \frac{\cos z}{z(z-3)}\,dz$

(d) $\displaystyle\int_{|z|=3} \frac{dz}{z^2+1}$

10. (Morera, primitives, and homotopy.)

(a) State Morera's theorem and the homotopy form of Cauchy's theorem.

(b) Let f_n be holomorphic on an open set U and suppose $f_n \to f$ uniformly on U. Prove that f is holomorphic on U.

(c) Show that $\int_\gamma z^{-2}\,dz = 0$ for every closed piecewise C^1 curve γ in $\mathbb{C}\setminus\{0\}$, but that the unit circle is *not* null-homotopic in $\mathbb{C}\setminus\{0\}$.